CYD EXTERNAL ANTENNA: THE 0-OHM RF SELECTOR RESISTOR MUST BE CHANGED
Move/resolder it from the PCB-antenna position to the IPEX/U.FL position.

Full Public BOM & Purchase Guide

ESP32-S3 Lite v1.1 | ESP32-WROOM v1.1 | CYD DEV Lite | CYD Expanded DEV

GitHub: https://github.com/beneltis21/BLE-Scanner

Price basis: eBay Australia only. Prices checked 26 August 2026. Prices, stock, selected variants and postage can change. Shipping is excluded from subtotals unless stated.

CRITICAL CYD HARDWARE REQUIREMENT

The CYD DEV Lite and CYD Expanded DEV external-antenna configurations require a CYD that actually contains an onboard U.FL/IPEX-capable radio connection.

The two development CYDs purchased from the linked vendor arrived with SparkleIoT ESP32 modules with onboard U.FL/IPEX connectors. The vendor listing photographs show a regular ESP32 module. The radio-module revision therefore appears variable/random, and another order from the same vendor is not guaranteed to contain the SparkleIoT/IPEX version.

MANDATORY HARDWARE MODIFICATION - 0-OHM RESISTOR

On the SparkleIoT/IPEX-equipped development boards, the external antenna connector is not automatically active simply because the socket is present.

The tiny 0-ohm RF antenna-selector resistor must be moved/resoldered from the PCB-antenna routing position to the U.FL/IPEX routing position.

If this resistor is not changed, attaching an external antenna does not switch the RF path to that antenna. The radio remains routed to the PCB antenna.

This resistor change is a required assembly step for the external-antenna CYD builds. Disconnect power before soldering. Fine soldering skill and magnification are strongly recommended. Identify the exact selector pads on the received board before moving the resistor.

Antenna assignment

- **CYD DEV Lite:** 3 dBi 2.4 GHz external antenna.
- **CYD Expanded DEV:** 8 dBi external antenna kit.
- Antenna gain does not guarantee a fixed detection distance.

Quick Cost Summary

Qty	Part	Price	Item / note
1 build	ESP32-S3 Lite v1.1 - board + short cable + removable mounting	AU$28.90	postage excluded
1 build	ESP32-WROOM v1.1 - board + LED + short cable + 3 wire colours + mounting	AU$40.97	postage excluded
1 build	CYD DEV Lite - CYD + 3 dBi antenna + 32 GB microSD + PSU + 1 m cable	AU$104.28	CYD postage excluded
1 system	CYD Expanded DEV - CYD + DB ESP32 + 8 dBi kit + SD + PSU + 2 cables + 3 wire colours	AU$160.22	postage excluded

How to read the totals

- Subtotals are indicative item prices, not guaranteed checkout totals.
- Postage is excluded unless specifically stated.
- The CYD listing is retained because it is the vendor source used for the development boards, but the SparkleIoT/IPEX revision is not guaranteed.
- For CYD external-antenna builds, **both** the onboard U.FL/IPEX connector and the 0-ohm RF selector-resistor modification are mandatory.

Current source status

- ESP32-S3: direct eBay Australia listing.
- ESP32-WROOM: direct eBay Australia listing; select the required 1-piece Type-C board option.
- CYD: direct development-vendor listing, but the radio-module revision is variable.
- microSD: eBay Australia product page; verify the active seller and current price before checkout.

CYD buyer rule: receiving a board with the IPEX/U.FL connector is only step one. On the development SparkleIoT boards, the 0-ohm selector resistor still has to be moved to activate the external RF route.

1. ESP32-S3 Lite v1.1

Use: simplest portable, phone-powered detector. No soldering is normally required.

Qty	Part	Price	Item / note
1	ESP32-S3 development board	AU$19.90	155140170255
1	0.2 m USB-C to USB-C data/power cable - select 0.2 m C-to-C	AU$4.05	178428447290
1	3M Dual Lock SJ3550 - 10 cm option - removable phone mounting	AU$4.95	124749045539

Indicative S3 subtotal: AU$28.90 before postage

[ESP32-S3 board](https://www.ebay.com.au/itm/155140170255)
https://www.ebay.com.au/itm/155140170255

[0.2 m USB-C to USB-C cable](https://www.ebay.com.au/itm/178428447290)
https://www.ebay.com.au/itm/178428447290

[3M Dual Lock mounting](https://www.ebay.com.au/itm/124749045539)
https://www.ebay.com.au/itm/124749045539

2. ESP32-WROOM v1.1 - NeoPixel

Use: low-cost portable build. Current public branch is NeoPixel only; no OLED.

Qty	Part	Price	Item / note
1	ESP32 WROOM-32 Type-C development board - select 1-piece Type-C	AU$20.39	137294733376
1	WS2812B / NeoPixel LED - selected listing is a 2-pack	~AU$4.35	178171022132
1	0.2 m USB-C to USB-C data/power cable	AU$4.05	178428447290
3 x 1 m	22 AWG stranded hook-up wire - 1 m - red/black/green	AU$7.23	222515259282
1	3M Dual Lock SJ3550 - 10 cm option	AU$4.95	124749045539

Indicative WROOM subtotal: AU$40.97 before postage

Wiring: NeoPixel data is GPIO 4. Three wire colours make power, ground and data easier to distinguish.

[ESP32-WROOM Type-C board](https://www.ebay.com.au/itm/137294733376)
https://www.ebay.com.au/itm/137294733376

[WS2812B / NeoPixel LED](https://www.ebay.com.au/itm/178171022132)
https://www.ebay.com.au/itm/178171022132

[22 AWG hook-up wire](https://www.ebay.com.au/itm/222515259282)
https://www.ebay.com.au/itm/222515259282

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Move/resolder it from the PCB-antenna position to the IPEX/U.FL position.

Use: stationary/facility and development build with screen and SD logging.

Do not treat the IPEX connector as plug-and-play. The received board must have the SparkleIoT/IPEX-style hardware or equivalent, and the 0-ohm RF selector resistor must be moved to the U.FL/IPEX route before the external antenna is used.

Qty	Part	Price	Item / note
1	ESP32-2432S028R CYD + case - variable radio revision	~AU$29.29	186604216967
1	2.4 GHz 3 dBi antenna + U.FL/IPEX pigtail	AU$15.95	273658085560
1	SanDisk Ultra 32 GB microSDHC Class 10	~AU$35.05	eBay product 5006829081
1	20 W USB-C wall adapter - select adapter-only	AU$12.00	145690077272
1	1 m USB-C to USB-C cable	AU$11.99	317696083162

Indicative CYD DEV Lite subtotal: AU$104.28 before postage

Development-vendor note: the linked CYD was approximately AU$29.29 when checked and showed approximately AU$55.84 international postage. The two development units received from this vendor contained SparkleIoT ESP32 modules with onboard IPEX/U.FL. The listing image shows a normal ESP32 module, so future supply must be treated as variable.

Required external-antenna preparation

- Inspect the delivered board and confirm the onboard U.FL/IPEX socket is physically present.
- Locate the 0-ohm RF antenna-selector resistor on the exact board received.
- **Move/resolder the resistor from the PCB-antenna route to the U.FL/IPEX route.**
- Inspect the work for bridges and confirm the selector has not been left on the PCB-antenna path.
- Only then attach the 3 dBi U.FL/IPEX antenna.

CYD vendor reference - variable radio-module revision
https://www.ebay.com.au/itm/186604216967

3 dBi 2.4 GHz U.FL/IPEX antenna
https://www.ebay.com.au/itm/273658085560

32 GB microSD
https://www.ebay.com.au/p/5006829081

20 W wall adapter
https://www.ebay.com.au/itm/145690077272

1 m USB-C cable
https://www.ebay.com.au/itm/317696083162

Confirm the actual CYD USB connector before ordering the cable. Substitute the correct cable if a delivered revision uses a different USB connector.

CYD EXTERNAL ANTENNA: THE 0-OHM RF SELECTOR RESISTOR MUST BE CHANGED

Move/resolder it from the PCB-antenna position to the IPEX/U.FL position.

Use: highest-capability stationary/development configuration. This is a two-board system.

1 OF 2 -> ESP32-2432S028R CYD with external antenna
2 OF 2 -> separate classic ESP32-WROOM database coprocessor

The 8 dBi antenna is specifically assigned to CYD Expanded DEV. The CYD's 0-ohm RF selector resistor must be moved to the U.FL/IPEX route before the external antenna can work through that connector.

Qty	Part	Price	Item / note
1	ESP32-2432S028R CYD + case - variable radio revision	~AU$29.29	1 of 2 - CYD
1	ESP32 WROOM-32 Type-C development board	AU$20.39	2 of 2 - DB ESP32
1 kit	2 x 8 dBi antennas + 2 x 35 cm U.FL/IPEX pigtails	AU$32.28	398276624134
1	SanDisk Ultra 32 GB microSDHC Class 10	~AU$35.05	5006829081
1	20 W USB-C wall adapter	AU$12.00	145690077272
2	1 m USB-C to USB-C cable	AU$23.98	one per board if USB-C
3 x 1 m	22 AWG stranded hook-up wire - 1 m - TX/RX/GND	AU$7.23	222515259282

Indicative CYD Expanded DEV subtotal: AU$160.22 before postage

UART: CYD GPIO22 TX -> DB ESP GPIO16 RX; CYD GPIO27 RX <- DB ESP GPIO17 TX; common ground required; 460800 baud, 8N1.

Optional enclosure: the generic ABS project box below may be used for the database board or mounting. Select size only after measuring the finished assembly.

CYD vendor reference - variable SparkleIoT/IPEX stock
https://www.ebay.com.au/itm/186604216967

Separate ESP32-WROOM database board
https://www.ebay.com.au/itm/137294733376

8 dBi U.FL/IPEX antenna kit
https://www.ebay.com.au/itm/398276624134

22 AWG UART wire
https://www.ebay.com.au/itm/222515259282

Optional ABS project box
https://www.ebay.com.au/itm/147517156820

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Move/resolder it from the PCB-antenna position to the IPEX/U.FL position.

The feature to look for is the small circular U.FL/IPEX antenna socket on the ESP32 radio module. A standard PCB-antenna module may look very similar but has no external antenna socket.

EXTERNAL-ANTENNA EXAMPLE	STANDARD PCB-ANTENNA EXAMPLE
ESP32-WROOM-32U / 32UE style	**Standard WROOM PCB-antenna style**
Small circular U.FL/IPEX socket present.	No U.FL/IPEX socket.
Connector-style reference only. A bare module is not a drop-in replacement for a complete CYD.	Not suitable for this external-antenna CYD configuration without proper RF/module rework.

Development CYD - SparkleIoT variant

The two development CYDs received from the linked vendor used SparkleIoT ESP32 modules with onboard IPEX/U.FL connectors. This is not guaranteed for future orders because the listing photographs show a regular ESP32 module.

Mandatory RF selector routing

Factory / PCB-antenna route	Required external-antenna route
RF COMMON -> [0 ohm selector] -> PCB ANT U.FL/IPEX route not selected	RF COMMON -> [0 ohm selector] -> U.FL/IPEX PCB-antenna route no longer selected

The resistor is the antenna selector. On the development SparkleIoT boards, connecting an antenna without moving this resistor leaves the radio routed to the PCB antenna. The resistor must be physically moved/resoldered to the external U.FL/IPEX path.

Component placement can vary between revisions. This guide therefore does not assign one universal pad number or orientation. Identify the selector on the exact board received before soldering.

WROOM-32U / 32UE examples

- ESP32-WROOM-32U: external-antenna module variant with U.FL/IPEX-style connector.
- ESP32-WROOM-32UE: newer external-antenna module family with the same general connector concept.
- These are examples of the connector style; they are not direct CYD replacements unless the module is properly integrated/reworked.

Espressif reference: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32d_esp32-wroom-32u_datasheet_en.pdf

Complete Current eBay Australia Link List

Build	Part / clickable link	Price	eBay item
S3 Lite	ESP32-S3 development board	AU$19.90	155140170255
Portable	0.2 m USB-C to USB-C data/power cable	AU$4.05	178428447290
Portable	3M Dual Lock SJ3550 - 10 cm option	AU$4.95	124749045539
WROOM	ESP32 WROOM-32 Type-C development board	AU$20.39	137294733376
WROOM	WS2812B / NeoPixel LED	~AU$4.35	178171022132
WROOM / Expanded	22 AWG stranded hook-up wire - 1 m	AU$2.41	222515259282
CYD both	ESP32-2432S028R CYD + case - variable radio revision	~AU$29.29	186604216967
CYD Lite	2.4 GHz 3 dBi antenna + U.FL/IPEX pigtail	AU$15.95	273658085560
CYD Expanded	2 x 8 dBi antennas + 2 x 35 cm U.FL/IPEX pigtails	AU$32.28	398276624134
CYD both	SanDisk Ultra 32 GB microSDHC Class 10	~AU$35.05	5006829081
CYD both	20 W USB-C wall adapter	AU$12.00	145690077272
CYD both	1 m USB-C to USB-C cable	AU$11.99	317696083162
Optional	Generic ABS electronics project box	AU$12.06	147517156820

Current price notes

- The CYD listing was approximately AU$29.29 plus approximately AU$55.84 international postage when checked.
- The 22 AWG wire listing showed AU$2.41 per selected 1 m colour and separate international postage.
- The 20 W wall-adapter listing showed AU$12.00 for the adapter-only selection when checked.
- The 32 GB SanDisk microSD product page showed approximately AU$35.05 for the active primary listing when checked.
- The LED listing is recorded at approximately AU$4.35 for the selected 2-pack; verify the active variant and postage at checkout.

Selection checklist

- Use a USB data cable where programming or serial access is required; charging-only cables are not sufficient.
- For CYD builds, verify the actual USB connector type on the delivered hardware.
- For CYD external-antenna builds, verify the onboard IPEX/U.FL socket before ordering the antenna.
- **After confirming IPEX/U.FL, move the 0-ohm RF selector resistor to the external-antenna path.**

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Purchase disclaimer

Links are provided as component-sourcing references and are not endorsements of a seller. Listing titles, selected variants, photographs, prices, postage and stock can change. Always verify the exact selected option before ordering.

CYD compatibility is a two-step requirement:
1. The delivered CYD must actually contain the onboard SparkleIoT/IPEX-style external antenna hardware or an equivalent compatible radio configuration.
2. On the development SparkleIoT configuration, the 0-ohm antenna-selector resistor must then be moved/resoldered from the PCB-antenna route to the U.FL/IPEX route.

A connector that is physically present but still routed away by the selector resistor is not an active external-antenna configuration.

Radio / antenna note

A higher-gain antenna can change RF characteristics and effective radiated power. This project uses BLE active scanning, which may transmit scan-request traffic as part of normal BLE operation. Builders should use compliant hardware and ensure the complete module/antenna configuration is appropriate for the jurisdiction in which it is operated.

Antenna gain does not guarantee a fixed detection distance. Reception depends on the transmitting device, antenna placement and orientation, walls and people, interference, reflections and the surrounding RF environment.

Assembly safety

- Disconnect all power before changing the RF selector resistor.
- Use fine soldering equipment suitable for small SMD components.
- Magnification is strongly recommended.
- Do not guess the selector orientation from another board revision; inspect the exact board received.
- Inspect the finished solder joint for bridges before applying power.

Build assignment

- ESP32-S3 Lite v1.1 - portable phone-powered build.
- ESP32-WROOM v1.1 - low-cost portable NeoPixel build.
- CYD DEV Lite - stationary build, 3 dBi external antenna.
- CYD Expanded DEV - two-board stationary system, 8 dBi external antenna; 1 of 2 CYD, 2 of 2 database ESP32.